

Microbial Fuel Cell-Based Biological Oxygen Demand Sensors for Monitoring Wastewater: State-of-the-Art and Practical Applications

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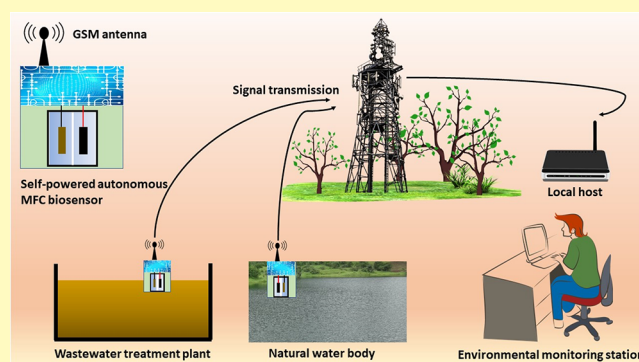
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ABSTRACT: Environmental pollution has been a continuous threat to sustainable development and global well-being. It has become a significant concern worldwide to combat the ecological crisis using low-cost innovative technologies. Biological oxygen demand (BOD) is a key indicator to comprehend the quality of water to guarantee environmental safety and human health; however, none of the present technologies are capable of online monitoring of the water at the source. Microbial fuel cells (MFC) are a promising technology for simultaneous power generation and wastewater treatment. MFCs have also been shown in fascinating applications to measure and detect the toxic pollutants present in wastewater. These are the bioreactors where exoelectrogenic microorganisms catalyze the conversion of the inherent chemical energy stored in organic compounds to electrical energy. Sensors employ energy conversion to measure BOD, which is considered an international index for the detection of organic material load present in wastewater. The MFC-based BOD sensors have gone through a wide range of advancement from mediator to mediator-less, double chamber to single-chamber, and large size to miniature. There have been detailed studies to improve the accuracy and reproducibility of the sensors for commercial applications. Additionally, multistage MFC-based BOD biosensors and miniature MFC–BOD sensors have also been ubiquitous in recent years. A considerable amount of work has been carried out to improve the performance of these devices by fabricating the proton exchange membranes and altering catalysts at the cathode. However, there remains a dearth for the fabrication of the devices in aspects like suitable microbes, proton exchange membranes, and cheaper catalysts for cathodes for effective real-time monitoring of wastewater. In this review, an extensive study has been carried out on various MFC-based BOD sensors. The efficiency and drawbacks associated with the different MFC-based BOD sensors have been critically evaluated, and future perspectives for their development have been investigated. The breadth of work compiled in this review will accelerate further research in MFC-based BOD biosensors. It will be of great importance to broad ranges of scientific research and industry.

KEYWORDS: microbial fuel cells, toxicity, biosensors, biological oxygen demand, direct electron transfer, online monitoring, Coulombic efficiency, dissolved oxygen



Water pollution has recently been addressed as a current environmental problem throughout the world since it stands as a major concern to human health.¹ Several techniques for the detection of the pollutants in wastewater have been put forth by various research groups.^{2,3} Since the off-site conducted chemical detection of contaminants is time-consuming, online monitoring systems have been prevalent in recent times.^{4,5} Among the diverse parameters used for monitoring different pollutants in wastewater, biological oxygen demand (BOD) is a vital parameter for understanding the quality of the wastewater.

According to International Union of Pure and Applied Chemistry (IUPAC), any analytical devices which can combine a physicochemical transducer including optical, magnetic, electrochemical, piezoelectric, and thermometric with an element that can identify the biologically derived molecules

such as antibodies, single-stranded DNA, enzymes, aptamers, and phages are defined as a biosensor.^{6–8} The abrupt alteration in the concentration of the target analytes or their sudden presence can be recognized by the bacteria. The process of recognition is sent as a signal and recorded as the output of the electric current.^{9–14} The biosensors are classified as enzymatic biosensors,¹⁵ immunosensors,¹⁶ biomimetic biosensors,¹⁷ DNA biosensors,¹⁸ and biosensors based on cells.¹⁹ It is

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worth mentioning that among the variety of biosensors, enzymatic biosensors have been of paramount significance as they are highly selective while recognizing their target molecules.²⁰ Nevertheless, certain limitations of enzymatic biosensors are the short half-life of the enzymes and the high cost of the enzyme purification and immobilization process.²¹ The whole-cell biosensors are considered as the best replacement of enzymatic biosensors.²²

The bioelectrochemical devices in which bacteria/electroactive microorganisms catalytically oxidize organic matter to produce electric current are termed microbial fuel cells (MFC).^{23,24} MFC were initially used for the production of energy from the renewable organic wastes present in wastewater.^{25,26} The chemical energy stored in organic compounds is converted into electrical energy through electrochemical reactions in MFCs.²⁷ The organic compounds are oxidized at the anode liberating electrons, flowing toward the cathode through an external circuit, thereby generating an electric current.⁹ The compounds present at the cathode, generally oxygen, accept the electron and get reduced to form water. The working principle of typical MFC is given in Figure 1.

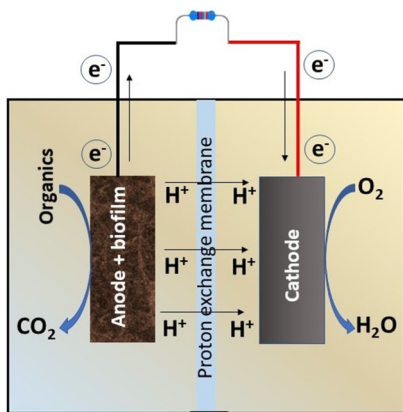


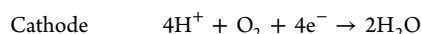
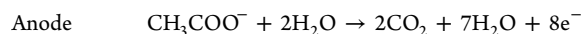
Figure 1. Schematic representation of typical microbial fuel cell and working principle. An electroactive biofilm on the anode oxidizes organic substrates generating electrons and protons in the process, adapted from refs 32 and 33.

MFC-based biosensors are devices that have been widely adapted for the monitoring of BOD based on its efficiency, ease of application, and economic viability as compared to the prevalent conventional methods.^{28,29} The detection of toxic materials in wastewater using MFC-based biosensors is in demand with progression of time, thus being among the most promising potential candidates of MFC technologies. Hence, these sensors have gradually developed from off-line to online monitoring systems.^{30–32}

MFCs are categorized as a type of biosensor in which the electrodes and proton-exchange membranes behave as the transducer, while the microorganisms present in the anode chamber act as an element for identification of biological compounds.^{34,35} The generated current is attributed to the respiration of the microbes, concentration of the fuels, and electroactivity of the anode breathing microbes. The distinctive assets of these MFC biosensors are exhibited by their capacity to perform in the absence of external power source or extra signal transducer, the anode biofilm being capable of innate self-healing in the absence of extra substrate/microbes and the

required organic substrates being the monitored water samples, which could not be achieved in the conventional biosensors.^{36,37} Recently, the MFC has been widely employed for analyzing different parameters like volatile fatty acids (VFA), chemical oxygen demand (COD), dissolved oxygen (DO), biological oxygen demand (BOD), and various toxic components present in wastewater.^{38–40}

The chemical reactions at anode and cathode are as shown below:



The biosensors based on the MFC are categorized into two types: the conventional double chamber MFC and single chamber MFC, which comprise an anodic chamber and air as the cathode.^{41,42} The basic components of a biosensor are bioreceptor, transducer, and signal processing system. The biosensor component, which consists of the elements for recognizing biological compounds, is designated as the bioreceptor. At the same time, the transducer converts the biological response to an optical, electrical, or thermal signal depending on the analytes' concentration.⁴³

Recent reviews on MFC-based biosensors describe different aspects like the detection of BOD, toxicity, and heavy metals.^{32,44} However, there has been a lack of review papers dealing specifically with the development of MFC devices as a BOD sensor and the production of electricity, which prompted us to conduct a detailed review focusing totally on this topic. The present review critically discusses the setbacks and arenas needed for the development in the MFC-based BOD biosensors for monitoring pollutants in wastewater. The detailed studies on the strategies to improve the accuracy and reproducibility of the sensors for commercial applications were thoroughly investigated. The efficiency and drawbacks of the different MFC-based BOD sensors have been critically evaluated, and future perspectives for their development have been investigated.

■ BIOLOGICAL OXYGEN DEMAND

The significant indicator for analyzing the capability of biodegradation of organic compounds in wastewater is signified as BOD. The amount of oxygen consumed by aerobic microorganisms for the degeneration of the organic compounds present in wastewater is calculated. The conventional procedure is the BOD₅-day test, which estimates the concentration of the dissolved oxygen in the sample by comparing the parameters before and after the completion of the 5 day incubation period at a temperature of 20 °C.⁴⁵ The BOD of a diluted sample is calculated using eq 1, as shown below:⁴⁶

$$\text{BOD} = \frac{\text{DO}_I - \text{DO}_F}{P} \quad (1)$$

where DO_I = concentration of the initial dissolved oxygen; DO_F = Concentration of the final dissolved oxygen; P = Decimal fraction of the bottle (300 mL).

The method mentioned above is popularly used but has its limitations of requiring long duration, labor intensity, and high maintenance cost. It is not practical for real-time monitoring and process control, which need spontaneous feedback. Recent trends have demonstrated the modification of biosensors based on dissolved oxygen probes, which have

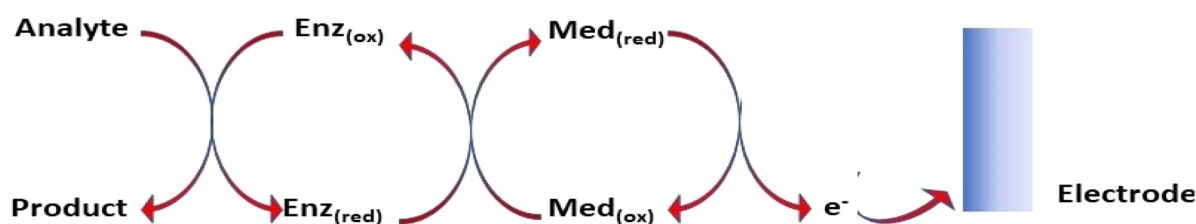


Figure 2. Schematic representation of mediated electron transfer; figure adapted from a previous study with permission from ref 56.

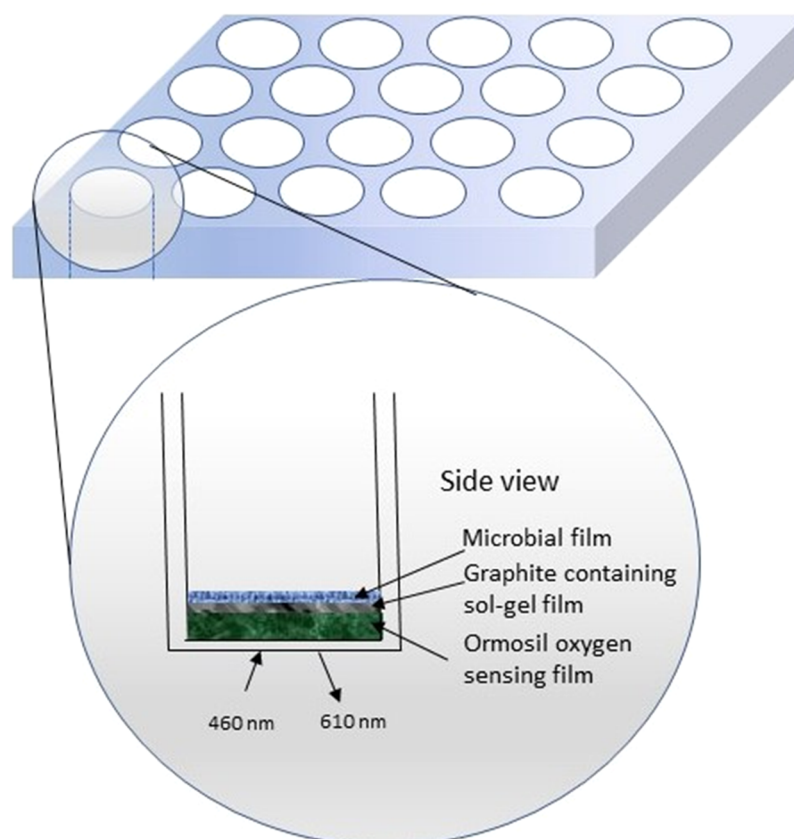


Figure 3. Schematic representation of optical BOD sensor; figure adapted from a previous study with permission from ref 64.

certain assets like working onsite, portability, extra processing steps like reagent addition, and small size.⁴⁷ However, these biosensors exhibited low stability over a long duration as the membrane becomes soiled over time. The substrate utilization is narrowed down in BOD sensors with a single organism. Additionally, several biodegradable chemicals affect the assessment of the BOD because of the fluorescence effect.⁴⁸ The stability of operation and range of measurements are weak for BOD sensors, which have further persuaded researchers to develop more efficient sensors.⁴⁹ The MFC-based BOD sensors have been found to exhibit stability in operation with low maintenance and exhibited promising technology as a BOD sensor.^{50,51}

■ VARIOUS TYPES OF BOD SENSORS

The main constituent of a biosensor is an element for biological recognition, which should be in direct contact with a transduction element.⁵² These devices use various transducers like colorimetric, electrochemical, luminescence, acoustic, and fluorescence transducers with different sensing elements like microorganisms, enzymes, receptors, antibodies, and nucleic

acids. Karube et al. reported the first BOD biosensor, which evolved later with a variety of modifications.

■ ELECTROCHEMICAL BOD SENSORS

The enzymes, whole cells, and antibodies are a type of electrochemical bioreceptor used to detect organic pollutants in wastewater. A Clark-type oxygen sensor is used for the detection of BOD in wastewater. The device consists of a biomembrane on which the microorganisms are immobilized along with an amperometric probe for analyzing the amount of dissolved oxygen.⁵³ An open-type electrochemical biosensor with an exposed anode was used for in situ detection of wastewater.⁵⁴ The concentration of BOD in wastewater was conducted by designing a membrane of cellulose acetate, which was immobilized with microorganisms. The device was equipped with an amperometric probe for oxygen made of gold or silver metal, which exhibited a detection limit of 1 mg L^{-1} and a response time of 7 min.⁵⁵

The movement of the oxygen between the atmosphere and the sample by diffusion is affected in Clark's electrode. The mediators for electron transport between the electrode and the

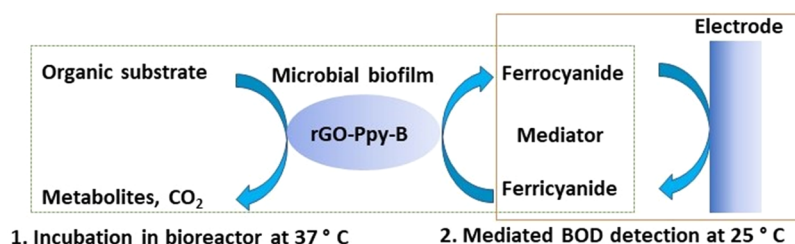


Figure 4. Schematic representation of the BOD detection mechanism by mediator type biosensor; figure adapted from a previous study from ref 70.

microbial cells are of low molecular weights. Thus, the thermodynamically feasible electron transfer is allowed, while the kinetically feasible electron transfer process is prevented. Therefore, the redox potential value of the mediator is a significant parameter for the transfer of electrons between the electrodes and the redox center of the cells, as shown in Figure 2.⁵⁶

■ OPTICAL BOD SENSORS

The optical sensors are associated with certain advantages like ease of fabrication, rapid response, lack of reference electrodes, no discharge of toxic chemicals, and no consumption of oxygen.⁵⁷ The intensity of emission of the fluorophores is dependent on the concentration of oxygen. Thus, the basic principle lies in the estimation of the fluorescence intensity of the fluorophores, which is reduced at the excited state by the oxygen molecule. The sensors for oxygen are prepared by using dichlorotris(1,10-phenanthroline)-ruthenium(II) dyes (a luminescent material), which are dissolved in Nafion.⁵⁸ Another approach is to design the sensor with bacteria, which is luminescent. The intensity of the respiration of the microbes is attributed to the amount of organic compounds in the sample, oxidized by microorganisms in aerobic conditions.

The BOD sensor using color *P. fluorescens* biovar V was used as a prokaryote microbe along with 2,6-dichlorophenolindophenol (DCIP), as a redox color indicator.⁵⁹ The deep blue color of DCIP was decolorized when reduced. The amount of dissolved organic compounds was proportional to the amount of the reduction of DCIP. A portable biosensing system (BODRCI-Pf) is equipped with a disposable chip of the optical cell, comprising three-cell units. Each cell constituted of DCIP and *P. fluorescens* biovar V cells.⁶⁰ It has three absorbance meters with a photodiode and a light-emitting diode. Another device was designed (BODRCI-Sc). The cell suspension was *S. cerevisiae*, which was mixed with DCIP and sample. The alteration in the absorbance was recorded before and after the incubation using a spectrophotometer. This method was found to be superior in the determination of BOD from seawater. Three fiber optic BODDO probes were developed for detecting seawater utilizing three different microbes: *Marinobacter marinus*,⁶¹ *Bacillus licheniformis*,⁶² and *Dietziamaris*,⁶³ respectively. Another study revealed the use of an oxygen sensing film on the fiber optic BODDO probe and modified it to an automatic system for sensing. Another probe was developed by coating a polystyrene microtiter plate with an oxygen-sensing film using a ruthenium dye and *Stenotrophomonas maltophilia* as the biosensing element as shown in Figure 3.⁶⁴

■ MEDIATOR TYPE BOD SENSOR

The conventional BOD sensors use microorganisms that are suspended in solution, demanding repeated centrifugation,

cultivating microorganisms, and washing. Thus, microorganisms are immobilized for obtaining improved stability and metabolic activity. The amount of dissolved oxygen in the sample will be affected by the respiration of the microbes, which is used for the analysis of BOD. Some specific microorganisms can reduce certain artificial electron acceptors⁶⁵ equipped with reversible redox properties. Thus, these mediator compounds can carry the electrons to the electrode from the biocatalytic microorganisms. Therefore, the analysis is independent of the partial pressure of oxygen within the medium. The mediator electrodes are independent of pH changes in cases where the reduced mediator's oxidation occurs without the participation of protons. The speed of analysis is higher, while a larger amount of current can be generated based on the large anode area in mediator-type BOD sensors.⁶⁶ The microorganisms have been immobilized in different geometrical patterns using cellulose,⁶⁷ agar,⁶⁸ and poly(vinyl alcohol).⁶⁹ The 3-dimensional graphene has also been used as mediators due to their improved properties attributed to the interconnected porous structure and π - π stacking of the graphene sheets.⁷⁰ These sensors have the possibility of being miniaturized, and they can also be used in the absence of oxygen since the enzymes can regenerate by reducing the artificial acceptors of electrons. Additionally, economic disposable biosensors can also be developed by utilizing screen-printed electrodes. The mechanism of detection is depicted in Figure 4.

■ BOD BIOSENSORS BASED ON THE OXYGEN ELECTRODE FILM

The BOD sensors constitute whole cells, and the amount of oxygen used by the microorganisms is analyzed; the microbes are immobilized on the transducer's surface. The microbial film layer that acts as the element for biological recognition is placed between the gas-permeable membrane and the porous membrane of the oxygen electrode. The amount of current generated is directly proportional to the amount of oxygen reduced at the electrodes. An equilibrium occurs between the rate of respiration of the immobilized microorganisms and the oxygen diffusion toward the immobilized microorganism layer. The organic substrates present in the wastewater are used by the immobilized microorganisms, thereby enhancing the respiration rate of the cells, which in turn augments oxygen consumption.⁷¹ Thus, the process depends on the rate of movement of the substrate to the immobilized layer.

The biosensor response methods for generating analytical signals are kinetic methods,⁷² where the relationship of strength of the current and time is used for a signal response. The equilibrium method using the response signal is based on the difference between the currents between both equilibrium states of the biosensor. However, the kinetic process is more acceptable among the two.

■ RESPIROMETRY BOD SENSORS

The respiration activity of the microbial suspension is measured with a respirometer, which in turn aids in determining the amount of BOD. Since the biorecognition element is not in contact with the transducer, this system cannot be strictly termed as a biosensor. The essential components of these systems are bioreactors containing activated sludge and organic substrates, which are easily oxidized.⁷³ The activated sludge acts as the element of biorecognition, and the transducer can be easily replaced. However, these systems are stationary and could not be used for direct measurements in the field.

■ PHOTOCATALYTIC BOD SENSOR

The photocatalytic reactor is composed of a reactor structure and a source of light. The output of the chemical reaction depends on the configuration and design of the reactor. The substrates are transported to the surface of the catalysts, where the photocatalyst degrades the organic pollutants. An efficient reactor should possess a large surface area for reaction, high kinetic rate, and speed of mass transfer.⁷⁴ However, the rate of photodegradation becomes very slow in the presence of low BOD concentration. The current output is enhanced with the increase in the value of pH. At lower pH, anionic electron donors and acceptors are favored, while at higher pH, the interaction between the cationic electron acceptors and donors is facilitated. The hydroxyl radicals initiate the decomposition of the organic compounds.

■ MICROORGANISMS PRESENT IN BOD SENSORS

Several microorganisms (*Torulopsis candida*, *Arxula adeninivorans*, *Escherichia coli*, *Serratia marcescens*, *Bacillus polymyxa*, *Issatchenkia*, *Klebsiella*, *Trichosporon*, *Hansenula anomala*, *Saccharomyces cerevisiae*, and *Bacillus subtilis*) have been used for designing BOD biosensors. The microorganisms are used as pure culture, mixtures of different pure cultures, or mixed cultures like BODSEED culture or activated sludge.⁷⁵ The single strain-based BOD sensors exhibit high stability and extended service life as compared to mixed cultures.⁷⁶ However, the narrow spectrum of the substrates of a single microbial strain decreases the value of the BOD-sensor.⁷⁷ The instability of the microbes can be hindered by killing the cells thermally, enhancing the biosensors' service life, as nutrient supply is not required for these cells.⁷⁸ In contrast, the living cells require nutrient supply and minerals along with maintenance. A BOD sensor's efficacy can be improved by using microorganisms that can absorb the refractory compounds present in wastewater, selecting microbes with large spectra of substrates. The bacteria can also be preincubated in the solution to be analyzed so that the cells can synthesize the required enzymes, which otherwise may not be produced.

However, these membrane-type BOD biosensors exhibit numerous limitations. These sensors fail to reproduce the results based on the lower amount of biomass immobilized on the membrane. They show poor stability, since there is resistance to the mass transfer by the microbial membrane on the sensor. At the same time, the DO and organic compounds permeate the membrane, decreasing the amount of DO. Therefore, the oxygen electrode should be highly sensitive.

■ PARAMETERS AFFECTING THE GENERATION OF CURRENT IN MFC BIOSENSORS

The generation of current in a microbial biosensor is affected by factors like pH, temperature, anode potential, and salinity.⁷⁹ The microbe's biocatalytic activity is directly related to the concentration of the organic matter added to the system under the condition that the parameters mentioned above remain constant in the case of nonsaturated organic substrates. The electrons are continuously transferred to the anode and stops when the organic matter becomes saturated. In the case of saturated organic compounds, the metabolic activity of the microbes and consumption of the substrates are restrained by different toxic compounds. This hindrance alters the generated current.⁸⁰ The Michaelis–Menten equation calculates the rate of inhibition (I) of the current due to the toxic substance present in the MFC biosensor,⁸¹ as shown in eq 2.

$$I(\%) = \frac{|CY_{\text{nor}} - CY_{\text{tox}}|}{CY} \times 100 \quad (2)$$

where CY = Coulombic yield in each peak which is analyzed by the integration of the electrical output; CY_{nor} = Coulombic yield in typical wastewater; CY_{tox} = Coulombic yield in toxic sample.

The Coulombic output is calculated by injecting a known concentration of toxic compounds into the anode chamber. The three standard toxic substrates are acetates (organic substrate), chromium (acute toxic compound), and iron (nontoxic metal).

The sensitivity of the MFC biosensor is defined as a change in the electrical signal per unit alteration of the concentration of the analyte. It is evaluated using eq 3, as shown below:

$$\text{sensitivity} = \frac{\Delta I}{\Delta c \cdot A} \quad (3)$$

where ΔI = unit change in current output; Δc = unit change in concentration of analyte; A = surface area of the electrode.

The organic substrates are used up by the assembly of bacteria to release the electrons at the anode. There is a potential difference between the equilibrium redox potential and the anode potential, termed the overpotential and analyzed by the Nernst equation, as presented in eq 4.⁸²

$$\eta = E_{\text{an}} - E_o + \frac{RT}{nF} \ln \frac{[\text{Ox}]}{[\text{Red}]} \quad (4)$$

where η = overpotential; E_{an} = anode potential; E_o = standard potential during the reaction; R = gas constant; T = temperature; N = number of electrons released; F = Faraday's constant.

There is a loss of energy at the anode due to toxic compounds, which produces a disturbance in the overpotential. The loss at the anode can be analyzed using a polarization curve and can be described by the Butler–Volmer Monod equation, as presented in eq 5.

$$I = I_{\text{max}} \cdot \frac{1 - e^{-\eta f}}{\beta_1 K_1 \exp[-(1 - \alpha)nf] + \beta_2 K_2 e^{-\eta f} + \beta_3 \frac{K_m}{s} + 1} \cdot \beta_4 \quad (5)$$

The sensitivity of a particularly toxic compound can be analyzed with the aid of this model since by noticing the alteration in the parameters, the outcome of the inhibition

kinetics of four different enzymes can be studied.⁸³ The toxicity can be detected online by analyzing the rate of consumption and concentration of the substrate, using the weighted least-squares process.

The electrochemically active microorganisms present in the anode compartment of MFC biosensors also behave as the biological recognizing elements. They generate the signal by responding to the toxic compounds at different concentrations and are inoculated by soil, anaerobic sludge, or domestic wastewater, which supplies an assortment of bacteria.^{84–86} Some reports have described the use of pure cultures at the anode chamber, which is uniform and highly stable.^{87–89} The use of a single strain of bacteria in the anode chamber of MFC biosensors has a vast prospect in future research endeavors. Another noteworthy parameter for the BOD sensor is the response time, which is defined as the time to obtain a fresh steady state after the BOD variation.⁹⁰ The factors influencing the response time are the cell structure, the rate of the fuel feed, and external resistance.

■ DIFFUSION OF OXYGEN

Unfortunately, the current output is found to be substantially decreased if there are strong electron acceptors like oxygen and nitrates within the chamber in the anode. These electron acceptors demonstrate intense competition for the electrons released from the biofilm.⁹¹ Although it can be eliminated by the addition of nitrate reductase (cyanide) and terminal oxidase (azide), these chemicals affected the cellular metabolism adversely by hindering the transfer of electrons through the extracellular membrane.^{92,93} The chemicals also required additional treatment before discharge to prevent the pollution of the environment. The effect of oxygen could be considerably reduced using a membrane that allows low permeability of gases. The stated obstacle needs to be addressed in future research prospects.

■ STABILITY OF OPERATION

The quality of performance of biosensors depends on their stability of operation. A stable biosensor should exhibit performance stability over a long duration of time. Concisely, the stability of operation depends on the microbes used, the sensor configuration, the immobilization method, the inner diffusion, and the external limiting substrate. The dissolved oxygen probe is less stable, imparting lower stability to the BOD sensors designed to monitor dissolved oxygen. The conventional Clark-type dissolved oxygen probe consists of a cathode (gold or platinum) and anode (silver), which tend to get oxidized easily,⁵³ demanding regular maintenance and cleaning. The major advantage of MFCBOD sensors is that they exhibit the desired operational stability.^{94,95} The MFC BOD sensors can oxidize a large amount of substrates as they consist of a large assembly of microorganisms.⁹⁶ However, the alteration in the composition of biofilm along with the biofouling of the proton exchange membranes with advancing time leads to a decrease in the activity of the sensors. The stability of the operation can be maintained by recalibration of the system at frequent intervals and reduction of the process of biofouling of the membranes.

■ EVOLUTION OF MFC AS BOD BIOSENSORS

Karube et al.⁹⁷ were the first to report the fabrication of biofuel microbial electrodes as BOD sensors. A pure culture of

Clostridium butyricum on the platinum electrode in the biofuel cell was used to estimate the BOD of wastewaters. Microbes were used for the transfer of the electrons, while hydrogen was utilized as the electron acceptor. A reduction in the generation of current was observed within 30–40 min, following which a steady state was obtained at 37 °C. The BOD of the wastewater was determined from the linear relationship between the BOD and the steady-state current. The analysis of the BOD estimation error was within a range of 10%.

■ MEDIATOR-LESS MFC-BASED BOD SENSORS

Although the substrate specificity is low, the MFC biosensors have provided massive success in BOD detection. The primary model of MFC was constructed using a double-chambered cell. However, various modifications of these devices were carried out to make it feasible for easy detection and economic viability. The MFC device that lacks mediators for transporting the electrons to the electrodes from the microbes produces higher potential than MFC equipped with mediators.⁹⁸ The main limitation of these MFC is that they should be fed with specific microbes. Kim et al.⁹⁹ reported a mediator-less fuel cell by employing an electrochemically active bacterium (*Shewanella putrefaciens*), which could reduce metals at the anode as a lactate sensor. The treatment of wastewater using the MFC-BOD biosensor revealed that the contaminants' strength in the wastewater was directly related to the current produced from the MFC.⁴⁹ Another mediator-less MFC was designed using oligotrophic microbes to improve the performance of the device.¹⁰⁰ Transparent polyacrylic materials were used for the construction of the cells along with oligotrophic microbes as the sensors. The size of the membranes which were selective to cations was decreased, and the platinum-coated graphite felt was used as cathode, while simple graphite felt was used as anode. It was concluded that the oligotrophic MFC produced low Coulombic yield, since the amount of oxygen generated by aerobic respiration for oxidizing the fuel at the anode was less than that of the oxygen diffusing to the anode. The capacity to transfer protons through the membrane was analyzed by reducing the membrane size, which in turn diminished the diffusion rate of oxygen, resulting in an enhanced Coulombic yield of the sensor. Therefore, by using both a membrane with a reduced size along with an enhanced catalytically active cathode, the amount of oxygen diffusing through the cation-specific membrane can be reduced, thereby improving the overall performance of MFC as low BOD sensor.⁹¹ The oligotroph-type MFC not only has high operational cost but also has excellent repeatability and reproducibility. An improved mediator-less MFC for continuous detection of BOD in wastewater with the aim of real-time monitoring was conducted using a model fitting study of the linear relationship of the BOD concentration.¹⁰¹ A solid line that was compatible with both lower and higher concentrations was derived in the study.

The operational ability of the mediator-less MFC-based BOD sensor was considerably enhanced through improving its dynamic response.¹⁰² An attempt was made to identify the input of the dynamic behavior of the MFC to the signal generated due to the output of current. The transitory response in an MFC by continuous checking of the BOD was analyzed. The consequences of the resistance applied to the external load and rate of fuel feeding on the sensitivity and the response time of the biosensor was noted. It was concluded that the external resistance and the hydraulic retention time

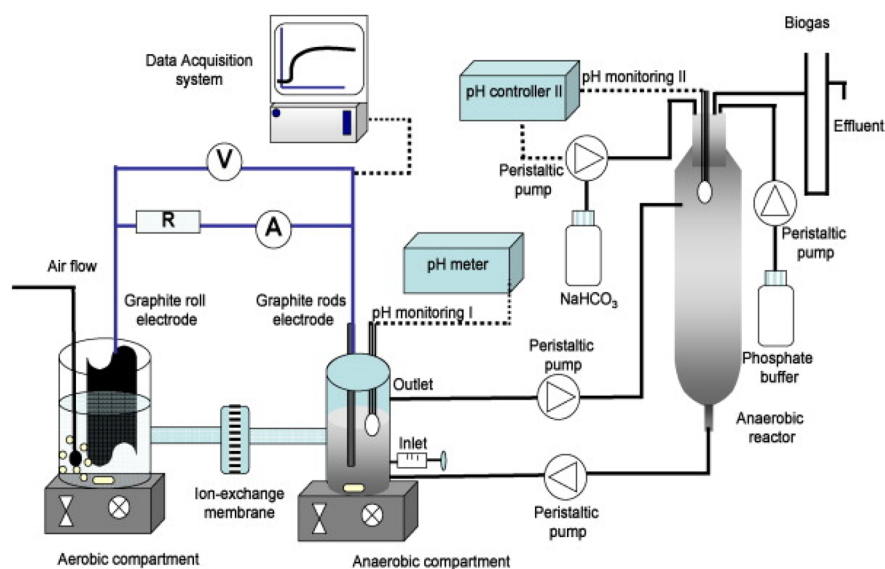


Figure 5. Schematic representation of the SCMFC BOD sensors; figure reprinted from a previous study with permission from ref 94.

impacted the dynamic operation of MFC. The current responded linearly to the alteration in the fuel concentration, ignoring the influence of electrochemical and microbial parameters subjected to the condition that the MFC is regarded as a plug flow reactor. The optimization of the rate of fuel feeding would lead to the new steady state at the particular hydraulic retention time. It was found that for a volume of fixed cell void, the response time could be lowered by reducing the hydraulic retention time. The S-type MFC and C-type MFC of varied heights revealed that the C-type MFC with low height was more efficient and scaled the MFC for decreasing the nonideal flow.

The removal of nitrate from the samples fed into MFC has continuously posed a serious problem in BOD sensors. It has been established that the primary enzyme of dissimilatory nitrate reduction called nitrate reductase and the terminal oxidase of aerobic respiration called cytochrome oxidase are restrained by compounds such as cyanide¹⁰³ and azides.¹⁰⁴ Accordingly, it was reported that quinol–cytochrome *b* oxidoreductase and coenzyme Q decreased the generation of current in MFC. Thus, the operation of MFC sensors was improved by manipulating the activity of the redox electron acceptors.¹⁰⁵ A low-intensity signal gets emitted from the MFC-BOD sensor in an environment containing high redox potential electron acceptors like nitrate and oxygen. This was because the generated electrons were not sufficient for producing the signal, as they were utilized by another reaction where substrates like oxygen and nitrate were reduced. These unwanted electron inhibitors were successfully removed by the addition of azide, cyanide, and rotenone to the cell without any prior treatment.

The detection capability of mixed microbe culture is efficient for many substrates. Henceforth, it is mandatory to identify each microbe present in a diverse culture to reproduce a similar MFC biosensor. A pure and mixed culture of microbes was used as biological sensing elements in MFC sensors while designing a mediator-less double chamber MFC-based BOD sensor by Hsieh and his group.¹⁰⁶ The assortment of bacteria (*T. carboxydiphila*, *P. aeruginosa*, *O. intermedium*, *S. frigidimarina*, *C. freundii*, and *C. acetobutylicum*) acting as biosensors was identified. An error rate within the range of 2.5–3.6% was

found while analyzing the BOD concentration in both domestic and polluted wastewater and confirmed the analysis of BOD concentrations up to a limit of 240 mg L⁻¹. The inoculated strains possessed the capacity to reduce metal and other complex compounds. Factors like effective operation, reproducibility, and high linearity made the device competent for commercial application. Another design of an artificial conglomerate of bacteria and complex natural bacteria from the activated sludge biofilm was used to detect BOD's concentration.¹⁰⁷ A detailed investigation of the structure of the bacteria was conducted through pyro-sequencing using the 16S rRNA gene, and the cluster of artificially designed bacteria like *Pseudomonas aeruginosa*, *Staphylococcus aureus*, and *Bacillus circulans* was identified on the anode. However, it was found that the natural bacteria were more efficient in detecting BOD concentration than that of the artificial assembly.

The advanced technology of the dual application of electricity generation from wastewater and its treatment using double chamber mediator-less MFC was reported¹⁰⁸ where a remarkable amount of electricity was generated, which aided in the effective detection of BOD. Another similar application was conducted to detect kitchen wastes by designing the system using two containers, fabricated with transparent polyacrylic plastic containers, eggshell as the proton exchange membrane, and electrodes of aluminum mesh.¹⁰⁹ There was a remarkable decrease in BOD concentration (54.55%) in molasses, while only 5.7% in the kitchen waste was observed.

■ SINGLE CHAMBER MFC-BASED BOD SENSORS

The main benefit of MFC with an air cathode is that it is devoid of processes like recycling, aeration, and regeneration of catholyte.¹¹⁰ The oxygen supply to the cathode gets enhanced, increasing the biosensors' sensitivity for a broad array of BOD concentration. A single chamber MFC with an anaerobic consortium was employed as BOD biosensor, where a stable conglomerate of anaerobic microbes was produced, for operating as an anaerobic reactor.⁹⁴ Graphite rods were used as the anode, while graphite rolls were used as a cathode, and the obtained data were recorded by the computer to produce the desired signals. The substrates were mixed inside the

anaerobic compartment, and the reactor system was designed like a cylindrical column, as illustrated in Figure 5. An anaerobic bioreactor was integrated for contributing a proper supply of anaerobic consortium, which provided good reproducibility of the results. The consortium's metabolic recovery took place simultaneously, which was regarded as the most significant asset of the system. However, the membrane had to be cleaned regularly, and there remained scope for enhancing the sensitivity by increasing the efficacy of the transport electrolyte.

In addition, the detection of labile organic materials in wastewater was pioneered⁹⁰ by designing a simple and compact single chamber microbial fuel cell (SCMFC) as a BOD biosensor with air as the cathode. This particular fuel cell was fabricated from polyacrylate tubes. The fuel cells had one flow port at the bottom while there were two others at the top. There were two sections (anode and cathode) at both ends of the tube. The cathode was made of carbon coating on platinum, which was exposed to air and separated from the anode by a cation exchange resin, while the anode was constituted of cloth made of carbon. The same group¹¹¹ further reported the improved performance of SCMFC-based BOD sensors by increasing the surface area of the carbon cloth anode. The anode was designed by employing a packed bed of granules constituted of carbon graphite. The design of the cell and configuration of the electrodes affected the operational capability of the MFC. The bulk solution's contact with the biofilm was efficiently improved with the manipulated design of the anode with a larger surface area. This modified design, in turn, led to a superior dynamic response of the system. It was determined that the surface area of the anode also affected the biofilm performance of the MFC system. A speedy response was exhibited by a pulse within a shorter time, along with short hydraulic retention time. The design of another single SCMFC–BOD sensor was conducted using cheaper materials like activated sludge and carbon felt as the anode for real-time BOD monitoring.¹¹² High activity of the electrochemically active bacteria present in the activated sludge for the online measurement of BOD systems was observed, with a simple operational process within a short duration of time. A standard deviation of the BOD values from 2.08% to 8.3% was found when examined for synthetic wastewater and real wastewater, proving the biosensor's practical efficiency.

The device was fabricated using a poly(methyl methacrylate) plate where the anode was made of PAN carbon graphite felt while Pt/carbon cloth was used as a cathode. An ion exchange film was used as a substitute for PEM, which considerably reduced the cost of the system. The external resistance and standard rate of flow with artificial wastewater was analyzed for establishing the optimal parameters for the best correlation between steady-state current and the concentration of BOD for artificial wastewater. A high linear response of 350 ppm of BOD concentration, and a better reproducibility (0.53%), was observed. Nevertheless, the response time was extended as compared to a double-chambered MFC. After conducting different optimization experiments for the rate of feeding of the fuel, the response time was decreased, and the steady-state current was attained. The system was modified by decreasing the distance between the electrodes, which reduced the resistance inside the reactor, thereby substantially enhancing the Coulombic efficiency due to the large surface area of the anode. A good correlation between the output of current and COD concentration for real wastewater was also obtained.

The composition of the electrode material, along with the reactor's configuration, mainly influences the capacity of power generation in a SCMFC device. Apart from this, the proton exchange membrane's activity, temperature, pH, and the source of substrates are also noteworthy factors for the current generation. The most commonly used material—Nafion—as proton exchange membrane had inevitable setbacks like high cost, loss of substrates, and permeation of oxygen from cathode to the anode, which finally produces an error in the BOD analysis. These limitations were surmounted by replacing Nafion with another polymer called sulfonated poly(ether ether ketone) (SPEEK).¹¹³ A comparative study about the activities of both proton exchange membranes, namely, Nafion and SPEEK, revealed higher Coulombic efficiency with SPEEK as compared to Nafion, since this proton exchange membrane enhances the rate of transport of proton from anode to cathode. The MFC consisted of a cylindrical chamber composed of acrylic and separated by SPEEK as a proton exchange membrane. The anode was a wet proof carbon cloth, while the cathode was also a carbon cloth coated with platinum catalysts. The sulfonation of SPEEK was enhanced to obtain equal conductivity rate of the proton as in Nafion. The desired anaerobic environment was also maintained in the anode by inhibiting the diffusion of oxygen from cathode to anode. The aerobic bacteria could not grow in the anode chamber due to the absence of oxygen, which prevented the loss of substrate. An amplified dynamic response was recorded, as the internal resistance demonstrated by SPEEK was low. This was ascribed to the higher amount of electron release by bacteria due to hindered permeability of oxygen through the membrane. An increase in the production rate of redox compounds and dynamic sensing of BOD compounds by 114% was found with SPEEK.

The enzymatic biosensors have been extensively used for detecting the sugar levels in the biological system since they are highly selective to a specific substrate. However, the main limitation of this process was that fuel cells fabricated by enzymes have short longevity ranging from days to weeks. Thus, these devices were not considered useful for online monitoring of different environmental issues, including wastewater monitoring. The design of self-powered SCMFC–BOD biosensors for online detection of wastewater was reported¹¹⁴ for the identification of urine as a contaminant in freshwater. It generated both visual and audio signals to the surrounding area. The online monitoring system was designed by the parallel electrical connection of four SCMFC. The cylinders for the MFC were made from terracotta cylinders, which also behaved as the cation exchange membranes. The anode was built by wrapping a carbon veil on the cylinder, while the cathode was designed by coating the carbon veil with activated carbon and PTFE. The cathodes were cut into rectangular pieces and placed inside the cylinder. The activated layer of carbon was kept in contact with the ceramic cylinder, while the water-repelling part of the carbon fibers was allowed to be in contact with the air. Acrylic plates were used for immersing half the portion of MFC in water, as depicted in Figure 6.

Seventy-five percent of the open-circuit voltage was applied as the final external load based on the assumption that the biofilm will be subjected to a long duration of the starvation period. This inhibited the catabolism of bacteria, thereby enhancing the time for signal transmission, which was an advantage for the system. The degeneration rate of biotic and abiotic elements was high in this environment under the

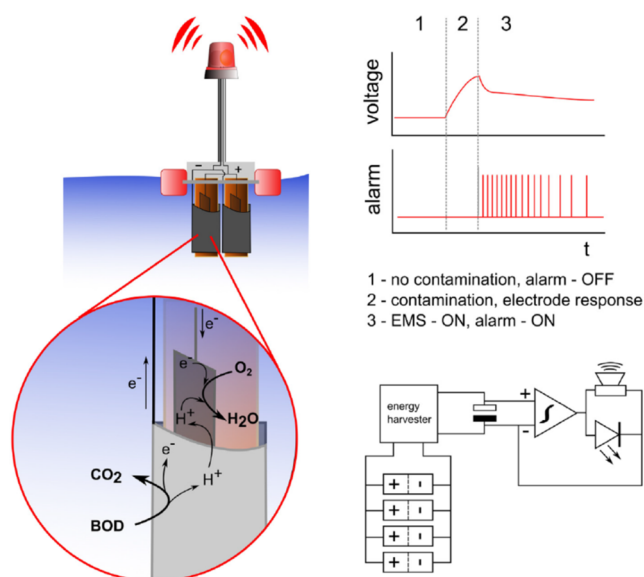


Figure 6. Schematic representation of the self-powered SCMFC BOD biosensors; figure reprinted from a previous study with permission from ref 114.

condition that the pollutant concentration was low. The external resistance could be manipulated to obtain a high-frequency signal. The open-circuit voltage of the MFC was increased on starting the sensor's open circuit mode, which subsequently initiated another cycle of the signal, implying that the electroactive bacteria were compatible with the lowering of urine concentration. The concentration of the urine was directly proportional to the frequency of the signal, which was reflected as an indicator, apart from the audio and visual indicating systems.

A cost-effective single-chambered MFC, equipped with MnO_2 as a catalyst for the air cathode,¹¹⁵ was used for the analysis of domestic wastewater and synthetic substrates for over the one-and-a-half-year duration. The fuel cell was fabricated with acrylic cylindrical reactors, where the conductive material was a mixture of carbon nanotubes and MnO_2 nanoparticles. The electrodes were made of carbon cloth, with a coat of MnO_2 on the cathode. The cathode was combined with Nafion as the proton exchange membrane. The substitution of expensive platinum catalysts with MnO_2 made the complete setup cheaper. The efficiency of the system was established by an excellent linear relationship in the case of both wastewater and sodium acetate between BOD and voltage.

The design of an inexpensive cathode using polytetrafluoroethylene and activated carbon in SCMFC as a BOD sensor for the first time reduced the cost of this device, since it replaced expensive Nafion and palladium catalysts.¹¹⁶ The anaerobic culture of microbes in activated sludge and synthetic wastewater was used in the device. The relation between the concentration of BOD and charge transfer was enhanced by high BOD concentration, which in turn increased the reaction time. The dynamic range of the sensor was modified due to activated carbon as the cathode material. Additionally, low resistance was offered by the cathode, since carbon black and stainless steel were good conductors. Moreover, the factor like the high specific surface area of activated carbon also contributed to the low resistance. Finally, polytetrafluoroethylene as cathode material was highly hydrophobic,

facilitating a high rate of oxygen diffusion, which diminished the system's resistance. The electrochemical activity of the air cathode also exhibited modified performance based on the low resistance. However, the biofouling of the electrodes caused a problem to the system.

■ SUBMERSIBLE MFC BOD SENSORS (IN SITU)

There are certain disadvantages with mediator-less MFC-based BOD sensors, such as difficulty setting up the system and inconveniences for in situ applications. The fabrication of a submersible microbial fuel cell (SUMFC) was designed to produce current from wastewater to improve the system.¹¹⁷ The design of the MFC was simplified by employing an air cathode. The anode electrode was immersed in an anaerobic reactor, and there was no requirement of an anode chamber.

Another submersible MFC as BOD sensor¹⁵ was fabricated with carbon paper and designed such that it was directly connected to the proton exchange membrane. The membrane was attached to the cathode (5% wet proofed carbon paper) by pressing in a hot condition. The air cathode surface was coated by a layer of platinum catalyst and attached to the proton exchange membrane. The electrodes were connected to resistors using wires coated with copper. The anode was directly submerged in anaerobic wastewater, while the cathode chamber was placed inside the chamber, where the air was sparged in continuously. Using this SUMFC, the analysis of the BOD values was found to be around 70 to 86 mg oxygen L^{-1} , which depends on the linear relation between the current density and the concentration of organic compounds present in the wastewater. The response time of the biosensor was lower than 10 h. However, pH and temperature affected the analysis of the current density. The biosensor could analyze the concentration of carbon within a low range to medium range. The positive feature of the system was the lack of requirement of the anaerobic chamber for the anode. Thus, the sensor could be directly immersed in the wastewater. Almost at the same time, another report¹¹⁸ documented the application of a submersible MFC for detecting the BOD identification using two different anodes: fresh anode or biofilm colonized by microbes as the anode. It was established that an anode colonized with biofilm was favorable for the analysis of BOD. In contrast, the sensor fabricated with fresh anode was perfect for evaluating the activity of the microbes. The analysis of both the concentration of microorganisms and the concentration of BOD was conducted from the linear relation with the density of current. A deviation of 6–16% for BOD was recorded, while 15–22% for microbial ATP concentration was reported. The same sensor could be ideally used for both types of anodes, and the sensitivity of the sensor was attributed to factors like initial pH, temperature, and conductivity of the system.

■ BIOELECTROCHEMICAL SYSTEMS

The production of electricity has been known by utilizing bioelectrochemical systems (BES) consisting of MFC.¹¹⁹ The biofilm formed over the anodes of the BES gets covered by exoelectrogenic bacteria (*Geobacter* spp.),¹²⁰ which combine with the anode, oxidize the acetate, and release the electron.¹²¹

A cheap bioelectrochemical cell with a modified BOD sensor was designed¹²² by replacing the ion exchange membrane with a filter paper which remained unaffected by alteration in pH. The biochemical reactor had three openings where two openings were for the effluent and influent liquids, while the

third opening was for the reference electrode. The anode was constructed from a graphite rod, and gas was diffused over the cathode, fabricated with filter paper made of carbon, and coated with carbon nanoparticles. The cathode which faces the liquid was also coated with polytetrafluoroethylene (30%), while the side facing the air was coated with polytetrafluoroethylene (40%). A Whatman filter paper was pressed on the cathode facing the liquid to act as a membrane. The conventional system of correlating the concentration of the BOD with the current was instead modified to be correlated with the amount of charge transferred. The efficiency of the sensor was enhanced by providing the feeds in batches. The resistor between the anode and the cathode was replaced by an external voltage, which improved current production. Furthermore, to avoid the pH shifts, which is not favorable for the sensor at low alkalinity, the ion exchange membrane was omitted entirely. The rate of reduction of oxygen by catalysts at the cathode and internal resistance of the cell generally impacts the current produced in conventional MFC. These aspects were considerably diminished by the input given as an external voltage. However, the applied voltage had to be limited as the water gets oxidized at a high voltage, which also gets added to the generated current. Nevertheless, the oxidation time of the microbes at high BOD concentration was comparatively higher, modified by increasing the ratio of the anodes' surface area to the volume. Another asset of this system was the insignificant surface area of the cathode, which was operated through applied voltage.

The creation of an in situ open-type BES-based BOD biosensor was designed⁵⁴ to be used directly in the intermediate aeration environment. The microbes of the biofilm formed on the anode were analyzed using 16S-rRNA genes through an upgraded sequencing method. Generally, the reaction in which electrons get transferred at the anode requires anaerobic conditions, and hence the MFC devices are so designed that the anode is placed within a closed chamber. However, the advantage of this specific biosensor is that it is a device devoid of a membrane, where the anode is the open type and can be directly dipped in water, as shown in Figure 7. The generation of current in this system during the aeration phase follows similar mechanisms as the conventional MFC

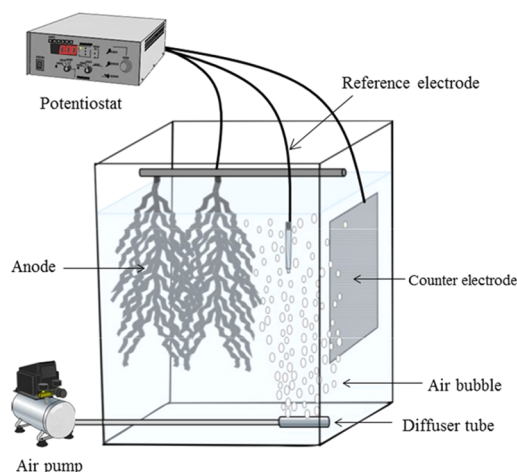


Figure 7. Schematic representation of an iBOB biosensor inside an aerating tank for continuous, amperometric BOD monitoring of wastewater; figure reprinted from a previous study with permission from ref 54.

devices. The presence of a thick biofilm on the anode facilitates the application of the anaerobic BES biodevices in aerobic systems. The presence of phylum *Planctomycetes* on the iBOB biosensor at high frequency, which constitutes anaerobic ammonium oxidation bacteria and aerobic heterotrophic bacteria, was already established.¹²³ A logarithmic correlation instead of the linear correlation of current with BOD was reported. Sensitivity up to 250 mg L⁻¹ was obtained, and the anode worked effectively without any maintenance as the attachment of the suspended solids benefited the anode in creating an anaerobic environment. In simple and conventional MFC, the current generation depends on both electrodes, while in an iBOB biosensor, the same depends solely on the anode, thereby producing stable current.

■ SCMFC AS NEURAL NETWORK

The complicated nonlinear relationship between the input and output data is recognized by mathematical models, known as artificial neural networks. These neural networks have the capability to identify intricate patterns and have been blended with an array of sensors for the identification of analytes and understanding the biosensor data.¹²⁴ Feng et al.¹²⁵ carried out the study using two conventional SCMFC equipped with a built-in anode, air cathode, and a cation exchange membrane, which was operated in batch mode. A study was conducted on four substrates: acetate and butyrate as nonfermentable substrates while cornstarch and glucose as fermentable chemicals. The identification between fermentable and nonfermentable substrates was recognized by applying the hyperbolic tangent sigmoid transfer function, as depicted in Figure 8. The injection of a single substrate produced four response metrics, which aided in recognizing the compounds present in the water sample. The testing of all four substrates randomly also recognized the peaks by using the four different hidden layers of neurons through continuous transfer functions. The information regarding the four-response metrics was transmitted by the application of the continuous transfer functions. However, the discontinuous transfer functions were not applicable for identification, a drawback associated with the system.

■ HYDRAULICALLY CONNECTED MULTISTAGE MFC

The anodic biofilm gets saturated with the increase in the concentration of the BOD. This condition leads to stagnant production of current as it causes a further rise in the concentration,⁴⁹ thereby hindering the monitoring of a large amount of organic compounds present in industrial wastewaters. To beat this impediment, a two-chamber MFC without a mediator was designed, which operated in three stages for treating artificial wastewater.¹²⁶ The structure of the consortium of microbes on the anode biofilms present in the continuous MFC system was quantitatively analyzed. The study was initiated with glucose as a fuel to understand the performance of the microbes present in biofilms used for the generation of electricity. The structure of the microbial community was studied by using molecular methods based on culture-independent 16S-rRNA gene. Three two-chambered MFC reactors were simultaneously connected through medium flow in series, each of which was equipped with Nafion as the proton exchange membrane. The anodes were made of porous carbon, while the cathode was made of porous carbon and platinum. The system was operated in a continuous

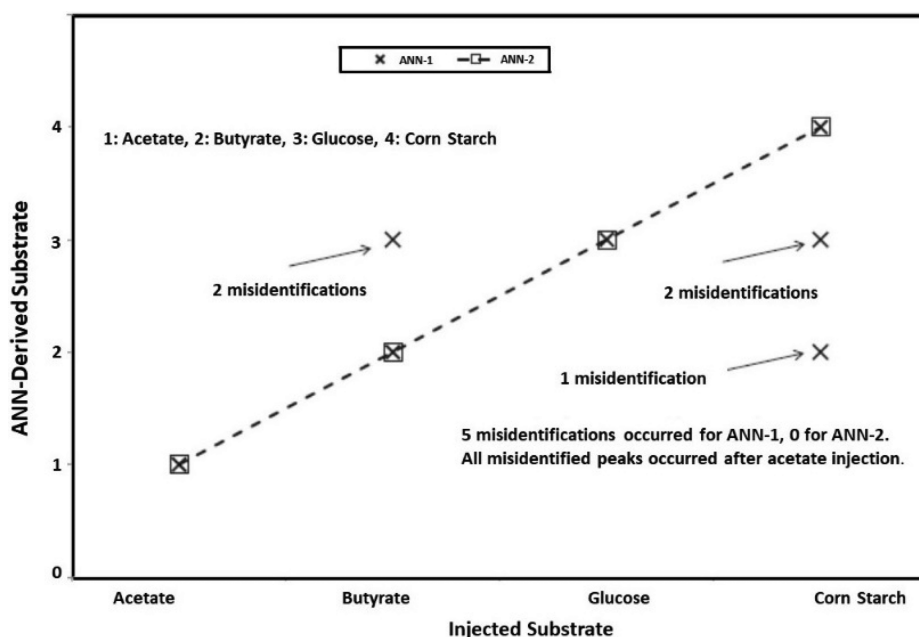


Figure 8. Artificial neural network result for four substrates; figure adapted from a previous study with permission from ref 125.

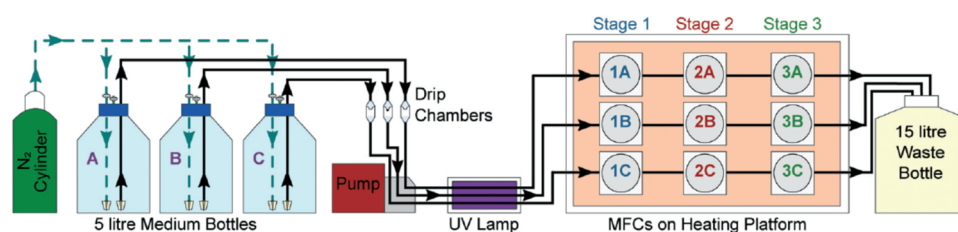


Figure 9. Schematic representation of the setup of three stages of MFC; figure reprinted from a previous study with permission from ref 127.

mode with a hydraulic retention time of 4.5 h. The effluent was transferred continuously from the first MFC to the second and subsequently to the third MFC. The microbial consortium present at the biofilm of the anode of first MFC was identified as anaerobic bacteria (*Aeromonas*), while Firmicutes was found at the anode of the second and third MFC. The value of the dissolved organic compounds removal was 85%.

Another multistage MFC system for enhancing the detection limit of BOD concentration was designed¹²⁷ using nine single-chambered MFCs, with carbon cloth as anode, carbon, and platinum as the cathode and Fumapem F-30 as cation exchange membrane. The continuous flow system was fabricated by using triplicate feed lines to the MFC, which were designed as a hydraulic collection in three stages, as presented in Figure 9.

The various specific microbes present in the biofilm were identified. The system was verified to establish the BOD detection limit and accounted for a much higher detection limit of BOD concentration (720 mg L^{-1}), as compared to the single-stage sensor BOD concentration. Apart from the enhanced detection limit of BOD concentration, another significant advantage of the multistage sensor is recognizing the cause behind the reduction of a signal. This reduction was attributed to either the generation of toxic chemicals or to diminishing concentration of BOD. The signal of the first stage MFC would be affected by the presence of toxic chemicals, while the signal in the final stage MFC was reduced due to the diminishing concentration of BOD. The range of biosensors

could even be manipulated to enhance the sensitivity of the BOD by improving the flow rate. Since the dynamic identification capability was not affected by saturated BOD concentrations at individual MFCs, the wastewaters with a high concentration of BOD could be easily detected using this multistage MFC.

The general MFC-based BOD sensors were incapable of identifying the elements (organic load or toxic materials), which caused a reduction in current. Thus, the monitoring of combined toxicity and BOD by utilizing a single sensor in multistage MFC was documented for improving the system.¹²⁸ The identification of the difference between inhibition due to non-biodegradable toxic compounds from reticence due to substrates was capable. The diminishing density of current in the third chamber indicated a decrease in the concentration of the BOD. In contrast, the excess concentration of BOD caused a loss of current in the first chamber. The presence of 4-nitrophenol as toxicant showed that the current was simultaneously inhibited from the first chamber to the last chamber as the compound moved through each of them. The identification of toxic organic compounds (4-nitrophenol) and control for the influence of the altering BOD concentrations using a multistage MFC toxicity sensor was also reported using a tubular-shaped multistage MFC equipped with four SCMFC, where the anodes were designed in a helical shape.¹²⁹ The population of the bacteria at the biofilm formed on the anode along with the production of current was decreased when the concentration of the toxic organic compound (4-nitrophenol)

was decreased. The production of current was influenced only by the concentration of the toxic organic compounds. There was a successive reduction of the current generation from the first stage to that of the last stage due to a considerable reduction of BOD concentration. Meanwhile, an equal reduction of current intensity was observed at each of the stages of MFC due to the presence of nondegradable toxic components. There would be an abrupt reduction of current in the first stage of MFC due to the presence of toxic compounds like 4-nitrophenol, which are easily degradable, while the generation of current would be subsequently enhanced in the next stages until it reaches the initial value of current production. Thus, the biodegradable capability of a toxic compound could be feasibly detected by this system.

■ MFC WITH EXCHANGEABLE MEMBRANES

The commodities like screws, clamps, or gaskets are not required in additive manufacturing techniques, which are also called 3D printing, providing the freedom to design different shapes using thermoplastic and flexible materials. The surfaces of the devices can be improved for making it porous and rough by increasing the thickness of the layer of print and also the resolution of the printer. This additive manufacturing technique catalyzes the synthesis of the biofilm and also removes the residual biomass. Zawadzki et al.¹³⁰ reported the design of a membrane for MFC using additive manufacturing techniques. A double-chambered MFC was designed, separated by an exchangeable membrane equipped with a movable module. It was applied for the detection of wastewater pollutants. Additive manufacturing techniques performed the synthesis of a biofilm in MFC. The MFC was fabricated with high-impact polystyrene, and the 3D-printed MFC was synthesized in organic solvents. The anode was fabricated of steel wool, which was placed in the anaerobic chamber while the source of microorganisms was sewage sludge for the anode. The aerobic chamber contained tap water with the agar membrane separating both chambers. The operation of the system was stable and reliable, opening the door for future research.

■ MEMBRANE ELECTRODE ASSEMBLY IN PROTON EXCHANGE MFC

The BOD sensor's performance was improved by the fabrication of membrane electrode assembly for proton exchange MFC.¹³¹ A carbon cloth and platinum-catalyzed carbon cloth were used as anode and cathode, respectively, separated by Nafion. The membrane electrode assembly was designed in the MFC by using *S. oneidensis* culture. The various assemblies of a membrane electrode were evaluated to find the appropriate ensemble. The reduction rate of oxygen on the cathode was enhanced by the contact between the electrodes, and the membrane is enhanced in this system. This is because the gap between electrodes and membranes is reduced in this design. The oxygen in the cathode reacts rapidly with the protons transferred from the anode, thereby exhausting the protons and improving electricity production. The combination of the air cathode with the membrane electrode assembly decreased the diffusion rate of oxygen to the anode, which prevented the reduction of electrochemically active bacteria. These factors were responsible for modifying the sensitivity of the BOD sensors.

■ MICRO MFC-BASED BOD SENSORS

Although numerous MFC-based BOD sensors have been developed, the low detection range of these sensors due to the slower response time had consistently posed a real challenge. Additionally, the response time does not seem to be enhanced even in the presence of a higher BOD concentration, which makes the devices ineffective for monitoring real wastewater. To improve the function of the sensors, a primary study on natural and artificial polymer membranes in miniature SCMFC for BOD sensors was done.¹³² The existing Nafion membrane in the SCMFC device was replaced by membrane fabricated from eggshell (natural polymer) and polydimethylsiloxane (artificial polymer). A comparative study using four MFC where two had polymer (eggshell and polydimethylsiloxane) as a membrane, while the third had Nafion as the membrane. The fourth MFC was a membrane-less device. The detection capabilities of the SCMFC with newly designed membranes were very high. The high detection capability was attributed to the enhanced ratio of the surface area to the volume of the electrodes due to the miniature size of the MFC. The transfer of mass was efficiently leading to the almost equal concentration of the substrates at the anode biofilm and bulk solution. The eggshell membrane and polydimethylsiloxane membrane after operation for 12 weeks indicated the deposition of a thick biofilm on the former while a thin biofilm on the latter. The formation of sporadic cells were also observed in Nafion. This indicates that different cultures of bacteria are present on membranes of different composition. The shape of the bacteria was rod-like in the newly experimented membranes, indicating the formation of a cluster of aerobic bacteria since eggshell and polydimethylsiloxane facilitate the permeation of oxygen through them. Thus, these membranes replicated a living membrane and decreased the oxygen in the anode, thereby enhancing the sensitivity. The eggshell membrane exhibits the lowest internal resistance and the highest sensitivity. In contrast, the maximum power density generated from the polydimethylsiloxane membrane was similar to that from a much more expensive Nafion membrane. Additionally, an increase in the power production was observed with a larger spacing of electrodes in the device with an eggshell membrane and membrane-less devices due to the reduction in oxygen cross-diffusion. At the same time, the same arrangement decreased the generation of power in the case of polydimethylsiloxane membranes. For these devices, the electrode spacing had a negligible effect on the BOD sensing capability.

The application of another micro MFC employing xerographic synthesis for BOD sensors was reported.¹³³ The system had the capability to integrate the ion transport membrane with different materials like platinum and carbon cloth for fabricating the electrodes. The device's performance with different response times was validated by operating in different modes: continuous flow mode and one sequential flow mode. Shorter response time with a high flow rate was observed, following the usual trend. The sensing capability was improved through biasing of the potential by a low voltage (1 mV) within a short response time. The internal resistance could be empowered through the process of potential bias, thereby allowing the bacteria to produce current with maximum output. Thus, the performance of the biofilm at the anode gets substantially powered, which in turn boosts the

activity of the BOD sensors.¹³⁴ A comparison table of BOD detection for different MFC is shown in Table 1.

Table 1. Detection of BOD with Various Types of MFC

serial no	type of MFC	substrate	BOD (mg L ⁻¹)	ref
1	Mediator-less MFC	Municipal wastewater	140	108
		Distillery wastewater	420	
		Dairy wastewater	270	
2	Single chamber MFC	GGA	1000	116
		Ethanol	320	
3	Submergible MFC	Groundwater	250	118
4	Bioelectrochemical reactor MFC	Acetate	640	122
5	Multistage MFC	GGA	1250	127
6	Miniature MFC	NaAc	723	133

■ COMPARATIVE STUDY ON MFC AND OTHER BIOSENSORS FOR THE DETECTION OF BOD

The type of immobilized microbial film determines the performance of a microbial BOD sensor based on its parameters like response time, range of response, service life, and sensitivity. The immobilization is carried out by different techniques like embedding, cross-linking, and absorbing¹³⁵ with carriers like agarose, Ca-alginate, etc. The dissolved oxygen and the organic compounds move into the biofilm through the porous structure of the carriers to react with the microorganisms, which in turn send the signal to the electrode. The porosity of these materials is not as required, which reduces the thickness of the biofilms as well as substrate dispersal on the film, thereby reducing the efficacy of the device. The system has been modified by altering the structure of the biosensors and improving the carriers.¹³⁶ Collagen fiber is a natural polymer that has been used on Zr(IV) to form ZrCF in which *E. coli* and *S. cerevisiae* microbes were immobilized to develop the BOD sensor for detecting wastewater, which was found to be 200 mg L⁻¹.¹³⁷

The identification and analysis of analytes have been conducted by luminescence reduction, which occurs due to

the physiological responses of microbes, which are luminescent to toxic compounds. The luminescent genes and bacteria have shown great promise as sensing elements in the device used for analyzing environmental pollutants. The consumption of oxygen is linked to the emission of light by the enzymatic reactions. The luminescent genes *lux A-E* obtained from *Vibrio fischeri* was combined with *Escherichia coli* cells for designing a BOD sensing system to detect wastewater by designing an imaging system. This system comprises a photomultiplier and a charge-coupled device camera.¹³⁸ The luminous microorganism can be used as disposable sensors.¹³⁹ The onsite BOD detection was carried out by designing a sensor made of perforated acrylic chip and immobilizing the luminous cells on the holes of the chip.¹⁴⁰ BOD detection was found to be 294 mg L⁻¹.

BOD sensors' main requirement are microorganisms with high activity of biooxidation and less selectivity for a broad range of organic compounds. A good correlation between BOD₇ and sensor BOD was obtained by using semispecific microbes for analyzing BOD in industrial wastewater. The semispecific microorganisms biodegrade certain refractory compounds and some specific compounds of the microorganism. A semispecific bacteria *Aeromonas hydrophila* P69 was used as the sensor for detecting pollutants in wastewater generated from the meat factory,¹⁴¹ and the BOD detected was 1224 mg L⁻¹.

The bioreactors have been used to design rapid BOD sensors that can biodegrade a large range of substrates. The rapid online wastewater determination of BOD was conducted with a plug-flow tubular biofilm reactor employing microbial seeds, and the value was found to be 0.088 mg O₂ L⁻¹.¹⁴²

Computing methods like artificial intelligence have replaced traditional methods. The artificial neural network (ANN) is a soft computing method used in the treatment plant of wastewater.¹⁴³ Two groups of ANN are utilized to control the desired parameters based on the selected parameters. Three ANN methods were used for detecting the BOD in wastewater, and the maximum amount was found to be 500 mg L⁻¹.¹⁴⁴

Graphene has been used for enhancing the electrochemical response of biosensors due to its good electrical conductivity,

Table 2. Comparison of Different BOD Sensors with MFC-Based BOD Sensors Having a Similar Microbe

serial no.	type of sensor	microorganisms	types of wastewater	BOD (mg L ⁻¹)	ref
1	Biofilm reactor	Natural microorganisms	Artificial wastewater	250	149
2	Mediator-less MFC	Natural microorganisms	Artificial wastewater	200	96
3	Reactor type	Activated sludge	Municipal wastewater	129	69
4	Graphene/MB/Chi-BSA-AS cryogel	Activated sludge	Wastewater from rubber gloves factory	3500 O ₂	146
5	Tubular biofilm reactor	Activated sludge	Municipal wastewater	0.088 O ₂	142
6	Two-chamber MFC	Activated sludge	Chocolate industry wastewater	230	150
7	MFC	Activated sludge	wastewater	158	151
8	Two mediator system (ferrocene-NR)	<i>Yeast D. hansenii</i>	Real wastewater	19.5 O ₂	148
9	Two-chamber MFC graphene electrode	Yeast	Seawater	0.16 O ₂	152
10	Semi specific biosensor	<i>Aeromonas hydrophila</i> / <i>Pseudomonas fluorescens</i>	Industrial meat wastewater	1224	141
11	Two-chamber MFC	<i>Aeromonas hydrophila</i>	Wastewater from the textile plant	692	153
12	Luminous bacteria sensor	<i>lux A-E/Escherichia coli</i>	Industrial wastewater	294	140
13	Zr loaded collagen fiber	<i>Escherichia coli</i>	Wastewater	234	137
14	Two-chamber MFC	<i>Escherichia coli</i>	Wastewater	40,000	154

flexibility, mechanical strength, and large surface area.¹⁴⁵ Methylene blue is used as a redox mediator, which also interacts with graphene through π - π interaction. The supporting materials determine the stability of biosensing elements which are immobilized. Chitosan-albumin is used to synthesize a porous cryogel. Thus, a biocompatible biosensor was designed using chitosan-bovine serum albumin cryogel on which the activated sludge was supported.¹⁴⁶ Methylene blue was used as a mediator for transferring the electrons from activated sludge to the carbon electrode. The sensor detected a BOD limit of 3500 mg O₂ L⁻¹ in the wastewater of a rubber and gloves factory.

The electrons can be effectively transferred to the electrodes from eukaryotic cells by using a double mediator system. Thus, a two-mediator system (H₂O₂-vitamin K3) based optical biosensor was designed using yeast (*S. cerevisiae*).¹⁴⁷ The chemiluminescence property is used as a signal for the BOD detection by using mediators (ferrocene-neutral red) and *D. Hansenii* yeast cells for designing an amperometric graphite paste electrode.¹⁴⁸ The amount of BOD detected was 19.5 mg O₂ L⁻¹ in wastewater.

Another BOD sensor was reported in which the elements of biomembrane were replaced by beads made of immobilized microbial cells placed within a chamber.⁶⁹ The matrix of the natural and synthetic polymer is used for the immobilization of the cells. The microbial cells were immobilized on beads of poly(vinyl alcohol) (PVA) and were used for the detection of BOD in municipal wastewater and found to be 129 mg L⁻¹.

A biofilm reactor was filled with natural microorganisms to form a natural biofilm for rapid online detection of BOD in wastewater with a new calibration method by systematic validations.¹⁴⁹ The value of the wastewater was found to be 250 mg L⁻¹. Comparative value of BOD detection with different sensors with MFC is shown in Table 2.

■ CONCLUSION AND FUTURE PERSPECTIVES

The development of MFC-based biosensors during the last 20 years has been re-evaluated in this paper. The continuous development from double-cell MFC to single-cell MFC and, finally, the modern design of multistage MFC for powering the performances have been discussed in detail. The researchers have conducted various tests with artificial wastewater and real wastewater with their designed MFC-based biosensors. The studies carried out by different research groups on modification of the proton exchange membrane and replacement of Nafion with better membranes have also been discussed. The additive manufacturing of the membranes for improving the performances of the MFC-based devices has also been addressed. The progress in the arena of the submersible MFC-based biosensors for in situ real-time monitoring of wastewater has been discussed in detail.

Nafion membrane is still widely used as proton exchange membranes in different types of MFC-based BOD sensors. Although few reports have demonstrated their replacement with other cheaper materials, there is still a need for a better material as a substitute apart from polymeric materials. There is ample scope in this area, like employing earthenware material and ceramic materials as proton exchange membranes. This condition affects Nafion negatively, as the formation of a biofilm on membranes incapacitates the ion exchange and assists in the transfer of a proton.¹⁵⁵ It has been reported that the performance of earthenware material increased with time as it could preserve the analytes for a longer duration.¹⁵⁶ Thus,

these substances as membranes can be developed for increasing the sensitivity of MFC-based BOD sensors.

The clayware used as proton exchange membranes in MFC have shown effective results as COD sensors, which have further prospects of improvement by enhancing the proton's conductivity and reducing the thickness for limiting the diffusion of oxygen.¹⁵⁷ Hence, this MFC with a modified proton exchange membrane can be extended to be used as BOD sensors, which will substantially reduce the cost of fabrication of the system since clayware is cheaper as compared to Nafion.

The self-powered independent MFC-based BOD sensors are an asset for monitoring wastewaters in remote areas that face a shortage of electricity. However, the actuation time for these MFC devices during the detection of a low concentration of contaminants was very high.¹⁰⁶ This realm requires elaborate research to reduce the response time and a possible application of detecting real wastewater.

Although a multistage MFC-based BOD sensor can differentiate between toxic compounds and BOD, there are inevitable setbacks in this method. A particular toxicant cannot be identified at a deficient BOD concentration, since the electricity generated depends only on the concentration of BOD. Thus, a low concentration of BOD will generate very low voltage current, hindering the identification of toxicants. Moreover, the structural design for multistage MFC calls for further improvements to address the problems associated with the accumulation of sludge. The sensing capability can also be further enhanced by manipulating the ratio of the electrode's volume to its area where the monitoring would be conducted devoid of aeration or dilution. The multistage MFC is in a budding stage, and the factors like pH, conductivity, and dissolved oxygen along with the composition of BOD require specific attention for monitoring real wastewater.

In the membrane electrode assembly, the requirement for the removal of water after operating for a long duration was a necessity, as the presence of water in the membrane prohibits the diffusion of oxygen through it. There is a demand for improvement by designing modified MFC, which would automatically remove the liberated water, thereby boosting the membrane performance for a long duration. There is still much to be studied about the robustness of the assembly of membrane electrodes with different fuels and its wide range of potential practical applications. Along this line, there also lie enough gaps in the knowledge of the electrochemical features of different MFC when different resistances are applied.

The major bottleneck for the MFC-based BOD sensor is associated with a natural, uncontrolled biofilm that could result in inaccurate measurement of organic load in the long run of the operation. The electroactive biofilms play a vital role in the electrochemical response and correlation to the solution's organic load or pollutant concentration. It is well-known that the biofilm composition can be altered easily upon the extrinsic parameters of the operating conditions and microenvironment where the system is being installed.

In the author's opinion, 3D biofilm printing could attribute a highly precise electrochemical response and the profoundly reproducible system. Thus, 3D biofilm printing can prove to be a highly promising technology for the fabrication of reliable, reproducible, and accurate MFC-based biosensors.

The designs of the 3D printed membranes in MFC, which can be readily exchanged, have opened a novel route for the development of MFC-based BOD sensors. The size and

materials of the membranes can be manipulated in innumerable ways for finding a convenient combination along with adaptable parameters, which could power the performance of the device exceedingly. This technology has opened a path for designing 3D printed BOD sensors for effectively detecting contaminants in real wastewater and other polluted water sources.

A high amount of BOD in the wastewater can be effectively analyzed by reducing the volume of MFC-based BOD sensors to enhance response time. Thus, the prospect of the application of a miniature MFC-based BOD sensor is a promising venture. The micro MFC-based biosensor has been used to detect a considerable COD concentration arena, where the peak current could detect the ratio of the nutrients in water samples along with aerobic and electroactive bacteria. These types of sensors can be calibrated for analyzing the BOD concentration indirectly, in cases where the ratio of the consumable bacteria to that of the organic composition, which is nonconsumable, is constant.¹⁵⁸

The submersible MFC, as a BOD sensor, holds strong promise for future development. The stirring of the system exhibited a decrease in the generation of current by the fresh anode. The stirring decreased the formation of the layer of electrochemically active bacteria on the fresh anode. The presence of nitrate within the system also decreased the generation of current, since the electrons were used up in alternate parallel reactions for the reduction of the nitrates. Although this unwanted electron inhibitor can be removed by the addition of some compounds like cyanide and azides, these additives are toxic to the environment. Thus, there is a scope for the modification of the submersible MFC with wireless technology for real online monitoring of wastewater.

The practical application of MFC is feasible based on the development of an energy harvester with high power. The stacked MFC has been successful to some extent for enhancing power production. However, the limitations of stacked MFC are the requirement of expensive catalysts, membranes, and different outputs from each MFC unit, which complicates the system's operation. Thus, there is always a search for a SCMFC with high power output, which can be used for wireless transmission. An ultralow power energy harvester for MFC (UPEM) was designed with a power of 3.08 mW, increased the voltage to 3.3 V.¹⁵⁹ This system was used for transmission of data to long-range within short intervals, powering module of carbon dioxide sensor and wireless communication. Since the AS-MFC requires manual introduction of the substrate, the MFC devices placed in natural environments are appropriate for this system. The power has also been generated using plant microbial fuel cells (PMFC), which is a new and green technology for the generation of power through degradation of the plant roots. The active bacteria present in the roots aid in the electrochemical process.¹⁶⁰ Continuous power is generated by PMFC applying energy harvesting systems, which is stored in a supercapacitor.¹⁶¹ An application of wireless sensor networks (WSN) was designed by utilizing self-powered Internet of things (IoT) equipped with ultrapower sensor nodes (SN) and harvesting the energy from PMFC.¹⁶² The wireless operation was carried out through a battery-less process where the SN mediated the power to the network of sensors. A continuous DC current was supplied to the SN through the fabricated PMFC-based LoRa sensor node. There lies a considerable prospect of designing cheap MFC devices for the detection of wastewater and generating high voltage

current using UPEM. This current can be used to power various wireless sensors in the near future. Additionally, the PMFC-based LoRa sensor node can be used for wireless monitoring of various environmental pollutants and current generation.

Despite the recent developments of various MFC-based BOD sensors for online monitoring of wastewater,¹⁶³ there is ample scope for further advancement for a broader application of these technologies for monitoring real wastewater. The primary area for development is to increase the selective capability of the sensors to trace secondary pollutants. The future aspects of the devices are their fabrication by using cheaper membranes, electrodes, and chambers, which will make them economically viable. Additionally, the devices' design and operation should be simple, with an attempt to make it practically applicable to real water monitoring.

The reproducibility of MFC-based sensors is of prime importance since the response generated from the system is directly correlated to the detection of compounds of interest and its concentration. The MFC system entirely relies on bacterial kinetics, which is governed by several intrinsic factors of the system, such as biofilm composition, operating temperature, and working conditions. The 3D biofilm printing can address those issues in the future by making tailor-made electroactive biofilms with the high specificity of a compound of interest. The precision of the system is another challenge that is also associated with the system kinetics, and that can be addressed by making electrodes using 3D printing technology. The fabrication cost is another hurdle as MFCs require an expensive precious metal catalyst for oxygen reduction reaction at the cathode side. The overall fabrication cost can be reduced significantly by the development and substitution of expensive metal catalysts. Thus, the cost of mass production of the MFC-based sensors can ultimately be reduced in the near future.

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Notes

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VOCABULARY

Microbial fuel cells, Microbial fuel cells are electrochemical devices that convert chemical energy into electrical energy using electroactive biofilm.; **Biosensor,** An electroanalytical device that combines biological elements to form an electrical signal. Bioreceptors identify the analyte of interest by producing a quantifiable signal and finally processed and read by electronic displays.; **Biochemical oxygen demand,** Biochemical oxygen demand is defined as the amount of dissolved oxygen required by aerobic microbes to breakdown organic matter present in the sample at a particular temperature over a specific time.; **Coulombic efficiency,** Coulombic efficiency (faradaic efficiency) is the efficiency of total electrons recovered by the electrochemical system in terms of total electric current.; **Dissolved oxygen (DO),** The amount of oxygen is dissolved in the water. The oxygen available for the microorganism and aquatic life signifies the quality of water.

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